

packages/graph-core/src/__tests__/fingerprint.test.ts

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Overview

A new test file `packages/graph-core/src/__tests__/fingerprint.test.ts` (lines R1-427) has been added.

It exercises the fingerprint utilities `contentHash`, `extractFileFingerprint`, `compareFingerprints`, and `analyzeChanges`.

Mocks for `node:fs` are set up via `vi.mock("node:fs")` (lines 12-16) and typed access via `vi.mocked` (lines 20-21).

Key changes

- Imports added at R1-6: `vitest` helpers, `StructuralAnalysis` type, and fingerprint functions from `../fingerprint.js`.
- Additional import of `readFileSync` and `existsSync` from `node:fs` at R18.
- Test cases cover:
 - SHA-256 consistency and uniqueness of `contentHash` (lines 27-38).
 - Extraction of functions, classes, imports, exports, content hash, and line count in `extractFileFingerprint` (lines 40-125).
 - Detection of change levels (`NONE`, `COSMETIC`, `STRUCTURAL`) in `compareFingerprints` across structural modifications (lines 128-330).
 - Classification of new, deleted, unchanged, and unknown files in `analyzeChanges` (lines 332-427).
- Edge-case checks for non-mutating array handling and conservative classification when `hasStructuralAnalysis` is false (lines 270-329).

Impact

- Provides unit coverage for fingerprint logic, helping detect regressions in change detection.
- Centralizes test logic; future updates to fingerprint functions will be caught early.
- Tests execute locally and do not alter production code.

Risks & follow-ups

- Mock reliability: ensure `vi.mocked` types match stubs; otherwise assertions may silently pass.
- Test flakiness: `contentHash` relies on deterministic SHA-256; confirm no environment-specific variations.
- Coverage gaps: verify all public fingerprint functions are exercised; consider integration tests for real file trees.
- Future API changes: if `FingerprintStore` or `StructuralAnalysis` evolve, update test fixtures accordingly.